TAC Type-6 – Seismic Cascade Theory
Corrected Version
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NOT EMPIRICALLY VALIDATED

Document Status

Type: Theoretical derivation / not empirically validated

This corrected version clarifies notation, explicitly marks hypotheses, and unifies the β-dynamics formalism.

Scope & Motivation

Type-6 describes a hypothetical, implosive-cascading coupling mechanism with strong local feedbacks. The goal is a consistent derivation of steepness and coupling relations that could arise in seismic-like Self-Organized Criticality (SOC) regimes.

Key Question: Can climate-driven threshold crossings trigger synchronized geophysical responses via cubic-root jump dynamics?

Central Assumptions & Notation (Clarified)

1. Coupling-to-Noise Ratio as Steepness Driver

The fundamental UTAC steepness parameter emerges from:

$$\beta \sim \alpha \frac{J}{T}$$

where:

- J = coupling strength (interaction energy)
- T = noise/damping (thermal fluctuations, stochasticity)
- $\alpha \approx 2$ = geometric prefactor (from RG theory)

2. Dynamic Steepness Near Threshold

Near the system threshold Θ, we hypothesize:

$$\beta_{\text{dynamic}} \propto (R - \Theta)^{-1/3} \text{ for } R \to \Theta$$

Physical interpretation:

- 3D parameter space (R, Θ, β) undergoes volumetric compression near threshold
- Projection onto 1D threshold coordinate yields cube-root singularity
- As $R \to \Theta$, effective steepness $\beta \to \infty$ (divergence)

3. SOC-like Energy Releases

Geophysical systems (earthquakes, volcanic eruptions) follow potential-energy landscapes:

- Power-law magnitude distributions (Gutenberg-Richter)
- Spatiotemporal clustering (aftershock sequences)
- Long-range stress coupling across fault networks

These characteristics are consistent with $\beta \approx 4.6 \pm 0.8$ (empirically validated, UTAC v2.0).

Derivation Sketch (Compact)

Step 1: Free Energy with Cubic Term Dominance

Near threshold, assume free energy expansion:

$$F(R) \approx F_0 + a(R-\Theta) + b(R-\Theta)^2 + c(R-\Theta)^3 + \dots$$

For systems with strong coupling and weak damping ($J \gg T$), the cubic term dominates.

Step 2: Projection from 3D to 1D

The 3D coupling space (R, Θ, β) projects onto the 1D threshold coordinate:

$$\beta_{\text{eff}} \sim \frac{1}{(R-\Theta)^{1/3}}$$

Step 3: Steepness Divergence

As $R \to \Theta$:

$$\beta(R) \to \infty \text{(singularity)}$$

This creates temporary extreme β-spikes ($\beta \gg 15$) near threshold.

Step 4: Consistency with RG Fixed Points

- **High-dimensional informational systems** ($d \gg 4$): $\beta \approx 4.2$ (CCUC)
- **Strongly coupled physical/biological systems**: $\beta \approx 7\text{–}13$ (higher damping, material constraints)

- **Near-threshold regime**: $\beta_{\text{dynamic}} \gg \beta_{\text{static}}$ (cubic-root jump)

Testable Predictions

1. Scaling Breaks in Aftershock Sequences

Hypothesis: Aftershock sequences near local Θ values should exhibit:

- Deviations from Omori law (power-law decay)
- Accelerated clustering as $R \to \Theta$
- Steepness spikes in temporal $\beta(t)$ estimates

Data required: USGS earthquake catalog, high-resolution strain gauge data

2. Cross-β Correlations (Climate-Geophysics)

Hypothesis: Climate-driven stress changes (ice mass loss, ocean loading) correlate with regional seismicity increases.

Coupling mechanism:

- Isostatic rebound: $\Delta\sigma \approx 1$ MPa (sufficient to trigger M 6–7 earthquakes)
- Methane hydrate destabilization: submarine landslides, tsunami generation
- Volcanic pressure release: reduced overburden → increased eruption probability

Data required: GRACE satellite (ice mass), USGS seismic catalog, cross-correlation analysis

3. Paleo-Seismic Validation

Hypothesis: Rapid deglaciation periods (12,000–8,000 BCE) show increased M ¿ 6 earthquake frequency.

Data required: Ice core chronology, seismic sediment records, statistical time-series analysis

Clear Status Warning

Relationship to UTAC v2.0 Main Paper

- **Main Paper (empirical):** Domain-specific β-clustering validated across 78 systems
- **This Paper (theoretical):** Proposed mechanism for extreme β-spikes near $R \approx \Theta$

- **Connection:** If cubic-root dynamics are real, they would explain the high-β tail ($\beta > 12$) observed in neurodegenerative and some climate systems
- **Falsification:** If empirical $\beta(R)$ measurements near threshold show NO divergence, cubic-root hypothesis is rejected

Next Steps for Validation

1. **Data acquisition:** USGS catalog reanalysis, GRACE-seismicity correlation
2. **Statistical modeling:** Time-varying $\beta(t)$ estimation near known thresholds
3. **Paleo-seismic analysis:** Literature review + proxy data compilation
4. **Publication:** Only after Step 1–3 show positive signal

Companion to: UTAC v2.0 Main Paper (DOI: 10.5281/zenodo.17472834)
Status: Theoretical/Speculative | **Date:** November 2025